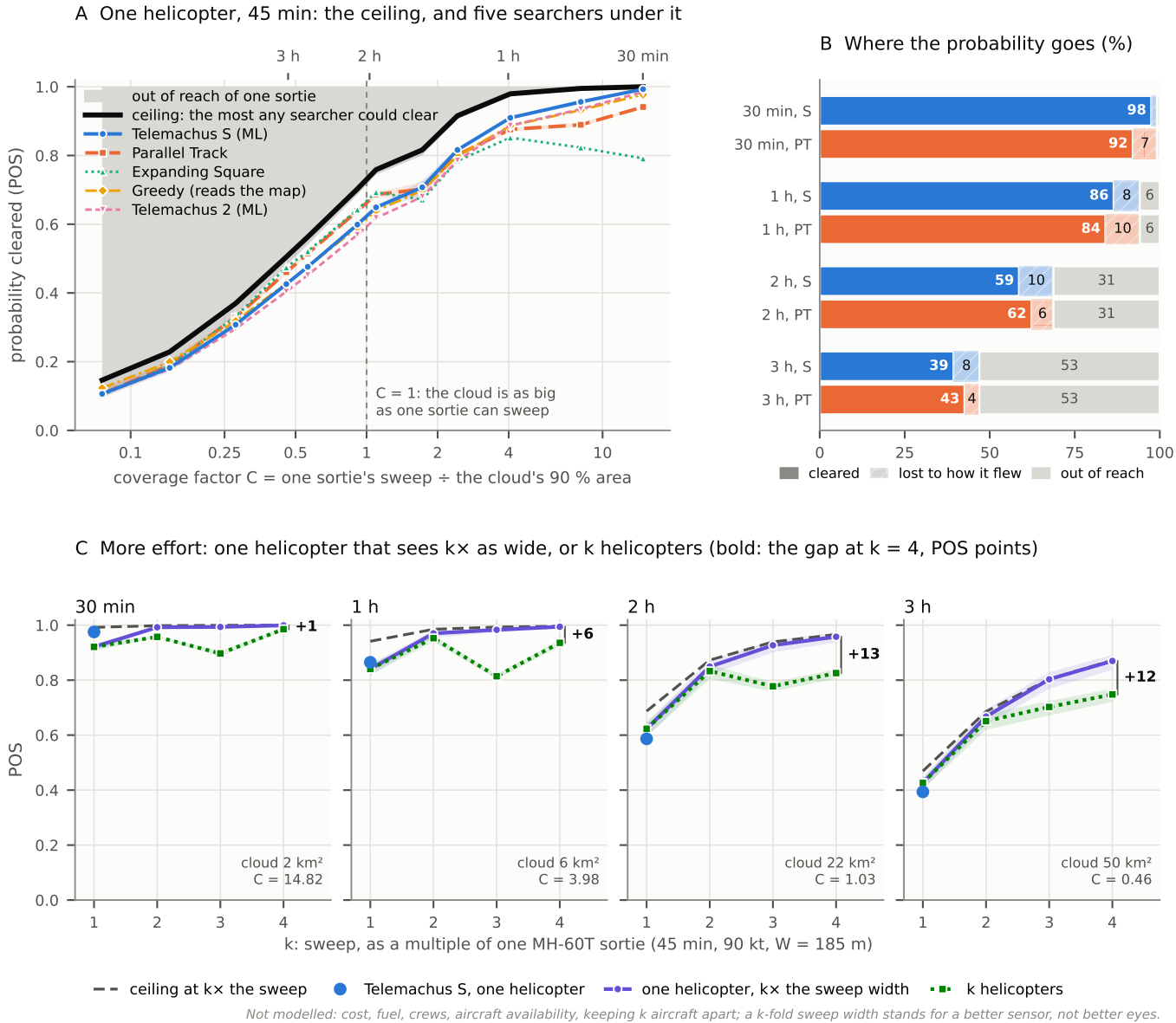
A One helicopter, 45 min: the ceiling, and five searchers under it
3 h
2 h
1 h
30 min
probability cleared (POS)
out of reach of one sortie
ceiling: the most any searcher could clear
Telemachus S (ML)
Parallel Track
Expanding Square
Greedy (reads the map)
Telemachus 2 (ML)
C = 1: the cloud is as big as one sortie can sweep
coverage factor C = one sortie's sweep ÷ the cloud's 90 % area
B Where the probability goes (%)
30 min, S
30 min, PT
1 h, S
1 h, PT
2 h, S
2 h, PT
3 h, S
3 h, PT
cleared
lost to how it flew
out of reach
C More effort: one helicopter that sees k× as wide, or k helicopters (bold: the gap at k = 4, POS points)
30 min
1 h
2 h
3 h
POS
+1
+6
+13
+12
cloud 2 km²
C = 14.82
cloud 6 km²
C = 3.98
cloud 22 km²
C = 1.03
cloud 50 km²
C = 0.46
k: sweep, as a multiple of one MH-60T sortie (45 min, 90 kt, W = 185 m)
ceiling at k× the sweep
Telemachus S, one helicopter
one helicopter, k× the sweep width
k helicopters
Not modelled: cost, fuel, crews, aircraft availability, keeping k aircraft apart; a k-fold sweep width stands for a better sensor, not better eyes.